

KATHLEEN CREEL

CONTACT INFORMATION

k.creel@northeastern.edu

WEBSITE

kathleenacreel.com

EDUCATION

University of Pittsburgh, Pittsburgh, PA

PhD in History and Philosophy of Science, 2020.

Dissertation: *Opening the Black Box: Explanation and Transparency in Machine Learning*

Robert Batterman (chair), Sandra Mitchell, James Woodard, Mazviita Chirimuuta, and David Danks.

Simon Fraser University, Burnaby, BC

Masters in Philosophy, 2015.

Williams College, Williamstown, MA

Bachelor of Arts in Computer Science and in Philosophy, 2010.

APPOINTMENTS

Northeastern University, Boston, MA

Assistant Professor, 2022 –

Department of Philosophy and Religion (50% appointment, tenure home),

Khoury College of Computer Sciences (50% appointment)

Stanford University, Stanford, CA

Embedded Ethics Postdoctoral Fellow, 2020 – 2022

Institute for Human-Centered Artificial Intelligence (HAI)

McCoy Family Center for Ethics in Society (EiS)

MIT Lincoln Laboratory, Lexington, MA

Software Engineer, Applied Research, 2010 - 2011

Group 64: Advanced Satellite Communications Systems

PUBLICATIONS

15. “Algorithmic Monocultures in Hiring,” *ACM FAccT 2026*. Written with Rishi Bommasani, Sarah H. Bana, Dan Jurafsky, and Percy Liang.
14. “Homogeneous Algorithms Can Reduce Competition in Personalized Pricing,” *NeurIPS 2025*. Written with Nathanael Jo, Manish Raghavan, and Ashia Wilson.
<https://arxiv.org/abs/2503.15634>
13. “Allocation Multiplicity: Evaluating the Promises of the Rashomon Set,” *ACM FAccT 2025*. Written with Shomik Jain, Margaret Wang, and Ashia Wilson.
<https://doi.org/10.1145/3715275.3732138>
12. “To Hedge or Not to Hedge: Scientific Claims and Public Justification,” *Philosophy of Science* (2024). Written with Zina B. Ward. <https://doi.org/10.1017/psa.2024.17>
11. “Ecosystem Graphs: The Social Footprint of Foundation Models,” *AI, Ethics, and Society* (2024). Written with Rishi Bommasani, Dilara Soylu, Thomas I. Liao, and Percy Liang.
<https://doi.org/10.1609/aies.v7i1.31629>
10. “Scarce Resource Allocations That Rely On Machine Learning Should Be Randomized,” *ICML 2024*. Written with Shomik Jain and Ashia Wilson.
<https://dl.acm.org/doi/10.5555/3692070.3692920>

9. “Algorithmic Pluralism: A Structural Approach To Equal Opportunity,” *ACM FAccT 2024*. Written with Shomik Jain, Vinith Suriyakumar, and Ashia Wilson. Winner of the FAccT 2024 Best Paper Award. <https://dl.acm.org/doi/10.1145/3630106.3658899>
8. “Ecosystem-level Analysis of Deployed Machine Learning Reveals Homogeneous Outcomes,” *NeurIPS 2023*. Written with Connor Toups, Rishi Bommasani, Sarah Bana, Dan Jurafsky, and Percy Liang. <https://dl.acm.org/doi/10.5555/3666122.3668350>
7. “Artificial Knowing Otherwise,” *Feminist Philosophy Quarterly* (2022). Written with Os Keyes. <https://doi.org/10.5206/fpq/2022.3/4.14313>
6. “Picking on the Same Person: Does Algorithmic Monoculture lead to Outcome Homogenization?” *NeurIPS 2022*. Written with Rishi Bommasani, Ananya Kumar, Dan Jurafsky, and Percy Liang. <https://dl.acm.org/doi/10.5555/3600270.3600535>
5. “The Algorithmic Leviathan: Fairness, Arbitrariness, and Opportunity,” *Canadian Journal of Philosophy* (2022) and *ACM FAccT 2021* (non-archival). Written with Deborah Hellman. <https://doi.org/10.1017/can.2022.3>
4. “Clinical decisions using AI must consider patient values,” *Nature Medicine* (2022). Written with Jonathan Birch, Abhinav Jha, and Anya Plutynski. <https://dx.doi.org/10.1038/s41591-021-01624-y>.
3. “Privacy and Paternalism: The Ethics of Student Data Collection,” *MIT Case Studies in Social and Ethical Responsibilities of Computing* (August 2022). Written with Tara Dixit. <https://doi.org/10.21428/2c646de5.b725319a>.
2. “On the Opportunities and Risks of Foundation Models,” *arXiv* (2021) Section lead for “Ethics of Scale”; written with Rishi Bommasani et. al., see full author list at <https://arxiv.org/abs/2108.07258>
1. “Transparency in Complex Computational Systems,” *Philosophy of Science*. October 2020, Volume 87 Issue 4. <https://doi.org/10.1086/709729>. Winner of the Philosophy of Science Association’s 2020 Ernest Nagel Early-Career Scholar Essay Award.

AWARDS

Best Paper Award, Fairness, Accountability, and Transparency (ACM FAccT), 2024.
Award recognizes the best six of 175 accepted papers.

Herbert A. Simon Award for Outstanding Research in Computing and Philosophy, International Association of Computing and Philosophy, 2023.
Award recognizes “scholars at an early stage of their academic career whose research is likely to reshape debates at the nexus of Computing and Philosophy.”

Ernest Nagel Early-Career Scholar Essay Award, Philosophy of Science Association, 2020.
Award recognizes “the best single-authored essay published in *Philosophy of Science* during the award year” by an early career scholar.

Elizabeth Baranger Teaching Award, University of Pittsburgh, Honorable Mention, 2019

Syllabus Award, competition organized by the International Society for the History Of Philosophy Of Science (HOPOS), 2018. Link to syllabus.

GRANTS

Principal Investigator (PI), amount \$400k, Alfred P. Sloan Foundation, “Cultivating Robust AI and Society Research Capacity and Community” (#2025-79205), 2025-2026.

Researcher, NSF Industry-University Cooperative Research Center, amount \$453k, “Phase 1: Northeastern University: Center on Responsible AI and Governance (CRAIG)”, PI John Basl, 2025.

Co-Principal Investigator, New America Foundation’s 2022 PIT-UN Network Challenge, amount (\$112k), “NORU-N Intelligence, Data, Ethics and Society Summer Institute”, PI Sina Fazelpour, 2023-2024.

Co-Principal Investigator, amount \$413k, NSF Research Community Development Grant, Summer Training Program to Expand the AI and Data Ethics Research Community (#2147220), PI Ron Sandler, 2022 - 2025.

Stanford Ethics, Society, and Technology Hub Award (\$15k), 2021

INVITED TALKS

Against the Platonic Representation Hypothesis

Popper Lecture, London School of Economics

Colloquium, Tufts University

Colloquium, University of Colorado, Boulder

May 2026

April 2026

February 2026

Randomness and Algorithmic Fairness

Colloquium, Brandeis University

Colloquium, Williams College

Colloquium, University of Wisconsin, Madison

Spring Colloquium, University of Michigan

Colloquium, University of Southern California

Center for Cognitive Science, Rutgers University

Colloquium, Utah State University

Seven Pines Workshop, Stillwater Minnesota

TOC4Fairness, Simons Collaboration, Berkeley

Munich Center for Mathematical Philosophy, Munich

Philosophy of Machine Learning, Tübingen

March 2026

November 2025

April 2025

April 2025

April 2025

December 2024

September 2024

May 2024

May 2024

January 2024

September 2023

A Molyneux Problem for Interpretable Machine Learning

Colloquium, Washington University, St. Louis

Colloquium, UCLA, History and Philosophy of Science Speakers Series

Interpretable ML in Natural & Social Sciences, Tübingen

Upstate Workshop on AI and Human Values, University of Rochester

March 2025

February 2025

October 2022

March 2021

Don’t Use Machine Learning to Evaluate Grants

Colloquium, Caltech, Pasadena

Orange Beach Epistemology Workshop, U. of Alabama

Colloquium, University of Leeds

Lingnan-Cambridge Conference on AI in Science

March 2025

May 2024

November 2023

November 2023

Machines - and Humans - Doing Science Together

Philosophy of Science: Past, Present, Future, Minnesota

October 2024

Homogeneous Outcomes in Algorithmic Hiring

Non-Market Effects of Market Power, Columbia

May 2024

Algorithmic Monoculture and Systemic Exclusion

Safra Center, Harvard University

Simon Fraser University, Vancouver

Symposium, AI and Society, UNC Chapel Hill

November 2024

November 2024

April 2024

Workshop, NIST, Northeastern	April 2024
Pecunian Society, Bowdoin College, Brunswick	April 2024
Colloquium, University of Utah, Salt Lake City	March 2024
Colloquium, Hamburg University, Germany	November 2023
Colloquium, University of Vermont, Burlington	October 2023
Colloquium, San Jose State University, San Jose	October 2023
Keynote, Hofstra University Presidential Scholars Symposium	September 2023
Keynote, International Association of Computing and Philosophy, Prague	June 2023
Workshop, Algorithmic Bias, Institute for Ethics in AI, Oxford	June 2023
Workshop, Ethical Foundations of Data Science, Rochester	April 2023
Workshop, Algorithms and Workplace Justice, Stanford	October 2022

Concepts on the Move: Transparency, Proxies, and Conceptual Engineering

Epistemological Issues of ML in Science, Dortmund Germany February 2024

Algorithmic Monoculture and Conceptual Change

Central APA, New Orleans February 2024

Embedded Ethics in Practice: An Integrated Approach to the CS Curriculum

FAIR AI, KAIST, Seoul June 2024
University of Utah March 2024

Trustworthiness in Medical AI

Data Science and Health, Vancouver November 2023

Picking on the Same Person: Homogenization and the Ethics of Scale

Cornell Tech, New York October 2022
Institute for Human-Centered Artificial Intelligence (HAI), Stanford May 2022
Pacific APA, Symposium on 'Attention' April 2022
HLRS High Performance Computing Center, Stuttgart April 2022
Machine Learning, Abstract Thought, & the Expanding Reach of AI:
Ethical and Conceptual Frontiers, University of Cincinnati April 2022
Colloquium, University of Wisconsin-Milwaukee March 2022
Google, Bioethics Speaker Series December 2021

Let's Ask The Patient: Stereotypes, Personalization, & Risk in Medical AI, July 2021

King's College London, Colloquium Series on Stereotyping and Medical AI

The Algorithmic Leviathan, co-authored with Deborah Hellman

Oxford University June 2022
Society for Practical Philosophy, keynote September 2021
University of Amsterdam, Netherlands February 2021
Institute for Human-Centered Artificial Intelligence (HAI), Stanford February 2021
Surrey Centre for Law & Philosophy: Symposium on Ethics & Algorithms January 2021
Workshop on Algorithmic Fairness, Copenhagen November 2020
TU Eindhoven, Netherlands June 2020

Function and User-Satisfaction in Explainable AI, ETAPS 2021

First International Workshop on Trusted Automated Decision-Making March 2021

Hermetica in Late Antiquity Workshop, Oslo

"The One and the Sun: Emanation and Hermetism in Plotinus" August 2019

What Makes a Good Explanation?

July 2019
Panel presentation in "What Makes a Good Explanation? Cognitive Dimensions of Explaining Intelligent Machines" at Cognitive Science, Montreal

REFEREED CONFERENCE PRESENTATIONS

The Political Role of Pluralistic Alignment Philosophy, Politics, and Economics (PPE)	November 2025
Conceptual Engineering and Large Language Models Philosophy of Science Association (PSA)	November 2024
Algorithmic Collusion from the (Normative) Ground Up Philosophy, Politics, and Economics (PPE)	November 2024
Ecosystem-level Analysis of Deployed Machine Learning NeurIPS, New Orleans	December 2023
Algorithmic Monoculture and Systemic Exclusion Oxford-Berlin Colloquium, Berlin	December 2023
Algorithmic Pluralism EAAMO, Boston	October 2023
Don't Use Machine Learning to Evaluate Grants British Society for Philosophy of Science	July 2023
Picking on the Same Person NeurIPS, New Orleans	December 2022
Machine Molyneux Problems Philosophy of Science Association, Pittsburgh	November 2022
To Hedge or Not to Hedge: Scientific Claims & Public Reason Values in Science and Political Philosophy, Claremont McKenna.	March 2022
A Molyneux Problem for Science Molyneux's Question: A Three-Dimensional Approach, Trinity College Dublin.	April 2021
Functional Approach to Explainable AI Overcoming Opacity in Machine Learning: Symposium at the Annual Convention of the Society for the Study of Artificial Intelligence and the Simulation of Behavior (AISB).	March 2021
Anti-Reductionist Machine Testimony Canadian Society for Epistemology, Montreal, Canada	November 2019
Noise in a Noisy World: Finding Metagenomic Patterns International Society for the History, Philosophy, & Social Studies of Biology, Oslo	July 2019
Early Modern Correspondences Teaching Hub Session: "New Approaches to the Early Modern Survey" American Philosophical Association Meeting, Pacific Division, Vancouver	April 2019
Demystifying Publishing Together: A 12 Week Graduate Workshop Minorities and Philosophy Session: "Creating Inclusive Spaces" American Philosophical Association Meeting, Pacific Division, Vancouver	April 2019
Signal, Noise, and the Detection of Scientific Phenomena American Philosophical Association Meeting, Central Division, Denver	February 2019

'There must be a Tub to Amuse the Whale' Joseph Black's Methodology Reconsidered History of Philosophy of Science Annual Meeting, Groningen, NL	July 2018
Explainable Artificial Intelligence Society for Philosophy of Science in Practice, Ghent	June 2018
Malebranche, Monsters, and the Maternal Imagination Newberry Library Multi-Disciplinary Conference, Chicago	January 2018
Transparency in Complex Computation Epistemic Opacity in Computer Simulation & Machine Learning, Stuttgart Models and Simulations 8, Columbia, SC Philosophy of Machine Learning Workshop, Irvine, CA Philosophical Perspectives on Data-Intensive Science, Hannover, DE	November 2018 March 2018 March 2018 October 2017
Machine Learning as Experiment , Pittsburgh Philosophy of Scientific Experimentation 4	April 2014

INVITED COMMENTARY

Northeastern Epistemology Workshop, Boston Commenter on "Uninformativeness as Unfairness," Zeynep Burçe Gümüşlü	April 2026
Book Workshop, Northeastern University Commenter on "What's Wrong With Stereotyping?," Erin Beeghly	March 2025
Southern Society for Philosophy and Psychology (SSPP), Cincinnati Author Meets Critics on "From Deep Learning to Rational Machines," Cameron Buckner	March 2024
Eastern APA, New York Comment on "Stereotype Engineering," Ian Olasov	January 2024
We Robot, Boston University Comment on "Taking Scale Seriously in Robotics and AI Law," Woodrow Hartzog and Mark McKenna	September 2023
Penn-Georgetown Digital Ethics Workshop Comment on "Value Encroachment on Scientific Understanding (in ML)," Emily Sullivan	March 2023
American Philosophical Association Meeting, Eastern Division Comment on "Just Machines," Clinton Castro	January 2021
American Philosophical Association Meeting, Central Division Comment on "Really Real Patterns," Tyler Millhouse	February 2020
Ethics of Technology At Work & in Public Institutions , Harvard Comment on "The Hidden Assumptions Behind Counterfactual Explanations & Principal Reasons," Barocas, Selbst, & Raghavan	Jan. 2020
Ethics of Algorithmic Decision-Making in Democratic Institutions Comment at Princeton on "Power, Process, and Automated Decision-Making," A. Waldman	Nov. 2019
American Philosophical Association Meeting, Pacific Division Comment on "Prediction in Science: How Big Data Does - and Doesn't - Help," R. Northcott	March 2018
Will Science Remain Human? , Rome Comment on panel of Paul Humphreys, Eric Winsberg, Emanuele Ratti	March 2018

INSTITUTES AND WORKSHOPS

Mentoring Workshop for Early Career Women* in Philosophy, 2025 WUSTL

A mentoring program for early career women* in philosophy, organized by the Marc Sanders Foundation.

Athena in Action

Summer 2018

Princeton, NJ

A mentoring program for graduate women in philosophy, organized by Jill North, Elisabeth Camp, and Elizabeth Harman.

Practices of Individuation and Classification in Science, June 2017 Banff, AB

Summer Institute organized by Alan Love, Ken Waters, Marcel Weber, and Bill Wimsatt as part of the “From Biological Practice to Scientific Metaphysics” initiative.

Workshop on Inclusive Pedagogy, February 2017

Pittsburgh, PA

Workshop on inclusive pedagogy by the American Association of Philosophy Teachers (AAPT).

TEACHING

Graduate Summer School

Northeastern University

2023-

- Co-taught an NSF-funded nine week summer school for advanced philosophy graduate students with Dr. John Basl. Organized and taught all aspects of the program, including:
 - Bi-weekly three hour philosophy of AI seminar (14 seminars/summer total).
 - Bi-weekly two hour computer science lab for philosophy grad students (12 sessions/summer total), including introduction to python, machine learning, sci-kit learn, neural networks, and language model architecture, culminating in a retrieval-augmented generation LLM final project.
 - Professional development workshop series, including topics such as Writing for Publication, Interdisciplinary Collaboration, and Preparing for the Job Market.
 - Organized visiting speaker series (10 speakers per summer).

Instructor of Record

Northeastern University

2022-

- **Knowledge in a Digital World** (2022, 2023, 2025): Explores epistemic questions related to digital technologies such as whether we can know an artificial system is conscious; whether we are in a simulation; how information should flow online and when such information is testimony; what constitute appropriate privacy norms; and how the use of machine learning and simulations has changed the epistemic practice of science.
- **Beyond the Binary: The Science of Race, Sex, and Sexuality** (2025): Considers how sex, race, and sexuality have been treated in science from the 19th to the 21st centuries. Examines the history of ideas about sex, gender, race, and sexuality as reflected in fields such as biology, anthropology, endocrinology, and neuroscience.
- **Philosophy of Science** (2023, 2024): Introduction to central topics in philosophy of science, including the nature of scientific method, scientific theories, scientific explanations, and values in science.

Embedded Ethics

Stanford University

2020-22

- Developed the first embedded ethics curriculum for the Stanford Computer Science department. Embedding ethical content in Stanford's introductory core computer science courses by delivering lectures, creating new lab assignments and ethics questions, and training section leaders.
- Courses include CS 106A: Programming Methodologies, CS106B: Programming Abstractions, CS107: Computer Organization & Systems, CS109: Probability for CS, CS161: Algorithms, CS221: Artificial Intelligence, CS234: Reinforcement Learning.

Instructor of Record

Williams College

2020

- **Ethics of Technology:** Explored ethical issues in machine learning and big data including transparency, surveillance, bias and fairness, automation and work, and the epistemology of deep fakes. Intensive one-month course meeting nine hours per week.

Instructor of Record

University of Pittsburgh

2017-2019

- **Problem Solving: How Science Works** (2019): Introduced students to problem solving, scientific methodology, and elementary probability and statistics.
- **Magic, Medicine, and Science** (2018): Mapped the roots and intertwined traditions of medicine, science, and alchemy in Europe and the Mediterranean 1550 to 1700, a period often called the scientific revolution. Special focus on the concepts of sympathy and force.
- **Morality and Medicine** (2017): Surveyed ethical issues in contemporary medical practice and research. Topics included informed consent; medical experimentation; race-based medicine; disability and wellbeing; and health care reform.

Teaching Assistant

University of Pittsburgh

2016

- **Philosophy of Science** (2016)

Teaching Assistant

Simon Fraser University

2011 - 2015

- **Introduction to Moral Philosophy** (Summer 2012, Summer 2013, Fall 2014)
- **Knowledge and Reality** (Fall 2011, Spring 2013, Fall 2013)
- **Critical Thinking** (Summer 2014)
- **Philosophy of Science** (Spring 2015)

Teaching Assistant

Williams College

2007 - 2010

- **Introduction to Computer Science** (2007-2010): Supervised lab help sessions and assisted professors 10 hrs per week in Java programming. Tutored students in Java. Teaching assistant for CS 134: Introduction to Computer Science and CS 136: Data Structures and Advanced Programming.

RELEVANT WORK EXPERIENCE

Software Engineer

MIT Lincoln Laboratory

2010 - 2011

Applied research on the Advanced Extremely High Frequency (AEHF) satellite constellation, a next generation secure satellite system, and its ground control terminals. Worked with a team to develop advanced high frequency waveforms, improve robustness of signal acquisition and tracking, and improve security. Maintained bug tracking database. Ran and developed test suites to ensure performance before deployment. Wrote code in C++.

Artificial Intelligence Research

University of Tennessee, Knoxville

Summer 2009

Wrote a control architecture in C++ for heterogeneous robot teams working with humans to perform tasks in an unknown environment as part of a research internship in Dr. Lynne Parker's Robotics and Distributed Intelligence Lab.

SERVICE

Dissertation Committee member:

- Sophia Wushanley (University of Michigan, Philosophy)
- Myraeka D'Leeuwen (UNC Chapel Hill, Philosophy)
- Conny Knieling (University of Pittsburgh, Philosophy)
- Shomik Jain (MIT, Institute for Data, Systems, and Society), PhD 2026
- David Liu (Northeastern University, Computer Science), PhD 2025
- Grace Fan (Northeastern University, Computer Science), PhD 2024

Undergraduate Thesis students co-advised:

- Marilyn Zhang (Stanford), advisor for "Affirmative Algorithms"
- Ryan Kearns (Stanford)

Reviewer for *Artificial Intelligence*, *British Journal of Philosophy of Science*, *Dialogue*, *Erkenntnis*, *European Journal for Philosophy of Science*, *Journal of Philosophy*, *Philosophy of Science*, *Philosophical Studies*, *Studies in History and Philosophy of Science*, *Synthese*, Dutch Research Council (NWO), NSF, *NeurIPS 2021-22 Datasets and Benchmarks Track*, *SIGSCE 2023*. Area Chair *FAccT 2025*, *2026*, *NeurIPS Position Track 2025*, *2026*, *ICML Position Track 2026*.. Program committee *AIES 2021-22*, *FAccT 2022 - 23*, *XAI 4 2022*. Panelist NSF ReD-DDoT, June 2024.

Presented at Philosophy in an Inclusive Key Summer Institute (PIKSI) for undergraduates, 2024, 2025

Presented at Intelligence, Data, Ethics, and Society (IDEAS) Summer School for undergraduates, 2023, 2024

Organizer, *AIDE Article Revision Group (AARG)*, 2024 -

- Led a twelve week article revision workshop for AIDE graduate students each fall.
- Helped graduate students to revise and polish papers written during AIDE Summer for publication by preparing weekly revision plans, leading writing support group, organizing draft exchanges, and commenting on manuscripts.

Founder, *Graduate Publication Workshop*, University of Pittsburgh, 2019-2020